

ASSIGNMENT NO. 3: Conditional Queries and Data Filtering

Questions:

Q1) Create a database named CollegeDB and a table named Student.

The table should contain the following columns:

Column Name	Data Type	Constraint
StudentID	INT	PRIMARY KEY
StudentName	VARCHAR(50)	
Age	INT	
Course	VARCHAR(30)	
City	VARCHAR(30)	
Marks	INT	
Email	VARCHAR(60)	

Insert at least 10 records into the table. Some records should contain NULL values in the Email column.

Query-

```
CREATE DATABASE CollegeDB;
```

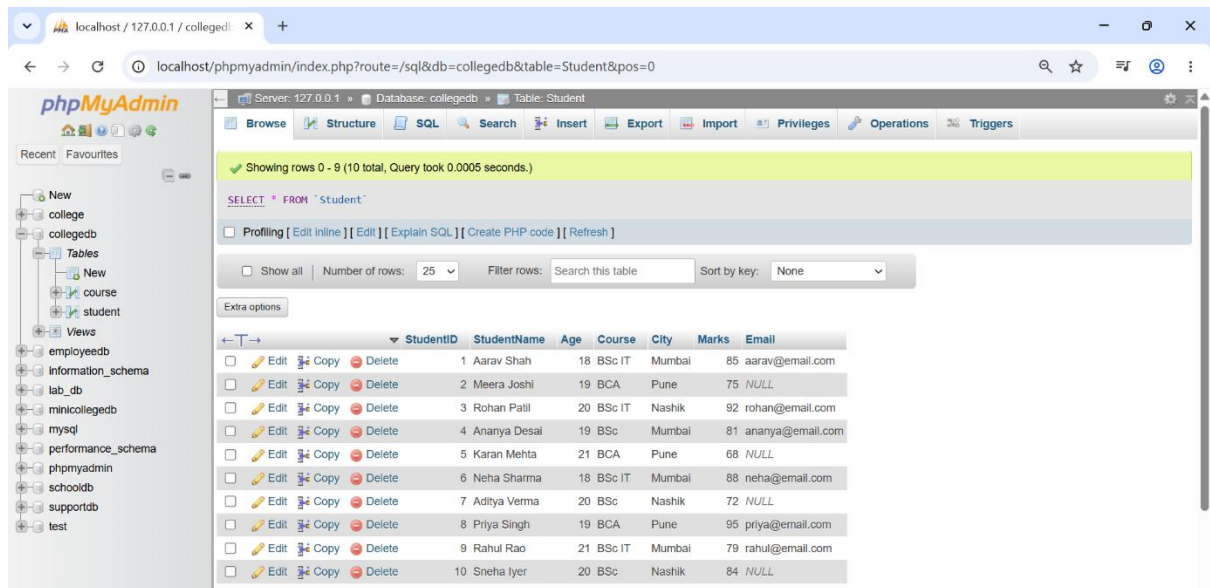
```
USE CollegeDB;
```

```
CREATE TABLE Student (  
StudentID INT PRIMARY KEY,  
StudentName VARCHAR(50),  
Age INT,  
Course VARCHAR(30),  
City VARCHAR(30),  
Marks INT,  
Email VARCHAR(60)
```

```
INSERT INTO Student (StudentID, StudentName, Age, Course, City,  
Marks, Email) VALUES
```

```
(1, 'Aarav Shah', 18, 'BSc IT', 'Mumbai', 85, 'aarav@email.com'),  
(2, 'Meera Joshi', 19, 'BCA', 'Pune', 75, NULL),  
(3, 'Rohan Patil', 20, 'BSc IT', 'Nashik', 92, 'rohan@email.com'),  
(4, 'Ananya Desai', 19, 'BSc', 'Mumbai', 81, 'ananya@email.com'),  
(5, 'Karan Mehta', 21, 'BCA', 'Pune', 68, NULL),  
(6, 'Neha Sharma', 18, 'BSc IT', 'Mumbai', 88, 'neha@email.com'),  
(7, 'Aditya Verma', 20, 'BSc', 'Nashik', 72, NULL),  
(8, 'Priya Singh', 19, 'BCA', 'Pune', 95, 'priya@email.com'),  
(9, 'Rahul Rao', 21, 'BSc IT', 'Mumbai', 79, 'rahul@email.com'),  
(10, 'Sneha Iyer', 20, 'BSc', 'Nashik', 84, NULL);
```

Output-



Q2) Write a MySQL query to display students who are from Mumbai and have marks greater than 80.

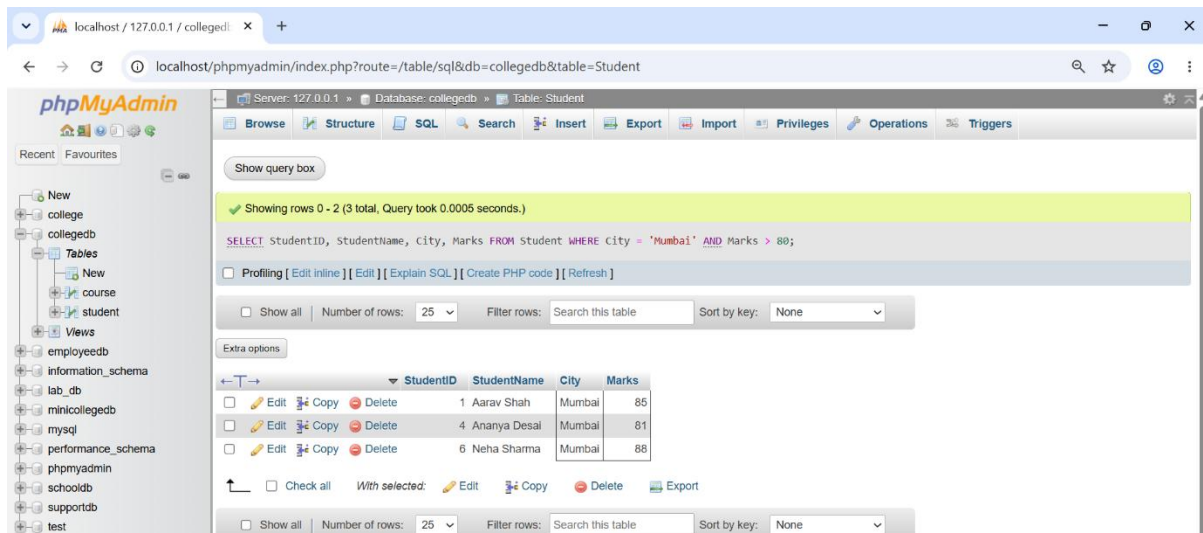
The output should include:

Column
StudentID
StudentName
City
Marks

Query-

```
SELECT StudentID,
        StudentName,
        City,
        Marks
FROM Student
WHERE City = 'Mumbai'
AND Marks > 80;
```

Output-



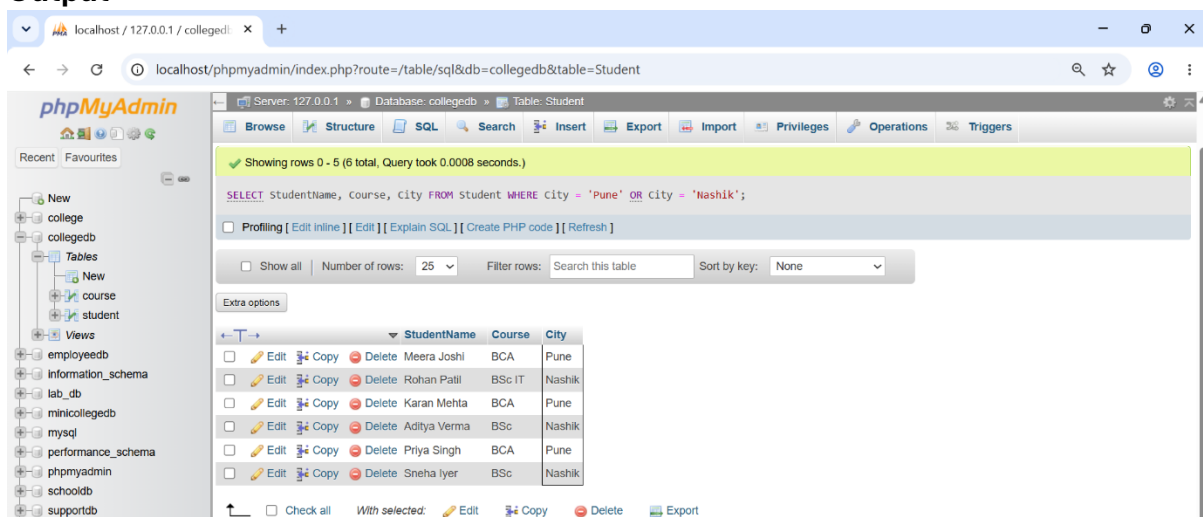
Q3) Write a MySQL query to display students who belong to either Pune or Nashik. Use the OR operator and display the following columns:

Column
StudentName
Course
City

Query-

```
SELECT StudentName, Course, City
FROM Student
WHERE City = 'Pune' OR City = 'Nashik';
```

Output-



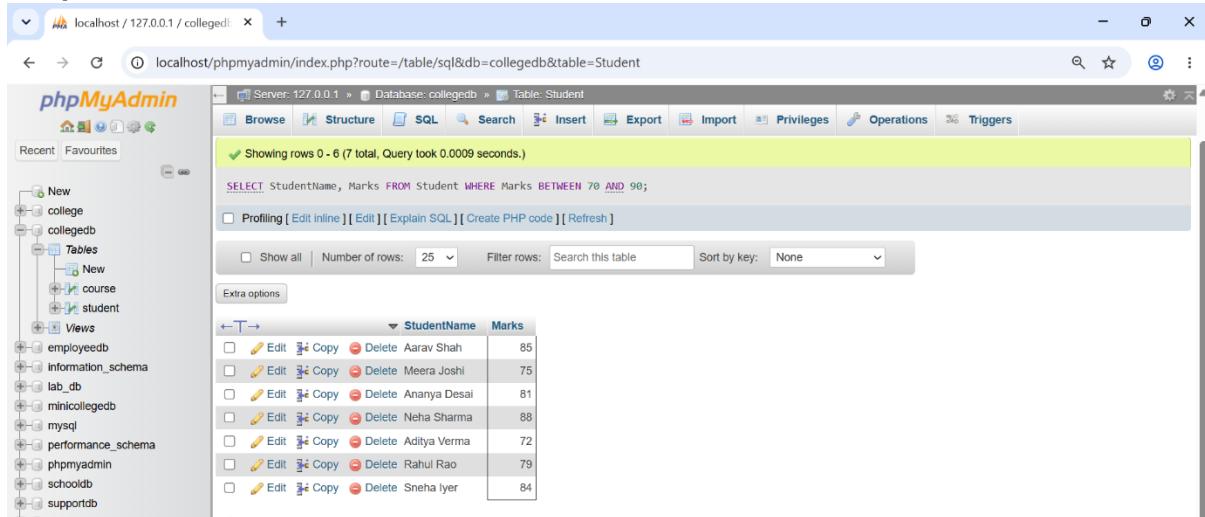
Q4) Write a MySQL query to display students whose marks are between 70 and 90. Use the BETWEEN operator and display:

Column
StudentName
Marks

Query-

```
SELECT StudentName, Marks  
FROM Student  
WHERE Marks BETWEEN 70 AND 90;
```

Output-



Showing rows 0 - 6 (7 total, Query took 0.0009 seconds.)

```
SELECT StudentName, Marks FROM Student WHERE Marks BETWEEN 70 AND 90;
```

Number of rows: 25 Filter rows: Search this table Sort by key: None

	StudentName	Marks
☐	Edit Copy Delete	Aarav Shah 85
☐	Edit Copy Delete	Meera Joshi 75
☐	Edit Copy Delete	Ananya Desai 81
☐	Edit Copy Delete	Neha Sharma 88
☐	Edit Copy Delete	Aditya Verma 72
☐	Edit Copy Delete	Rahul Rao 79
☐	Edit Copy Delete	Sneha Iyer 84

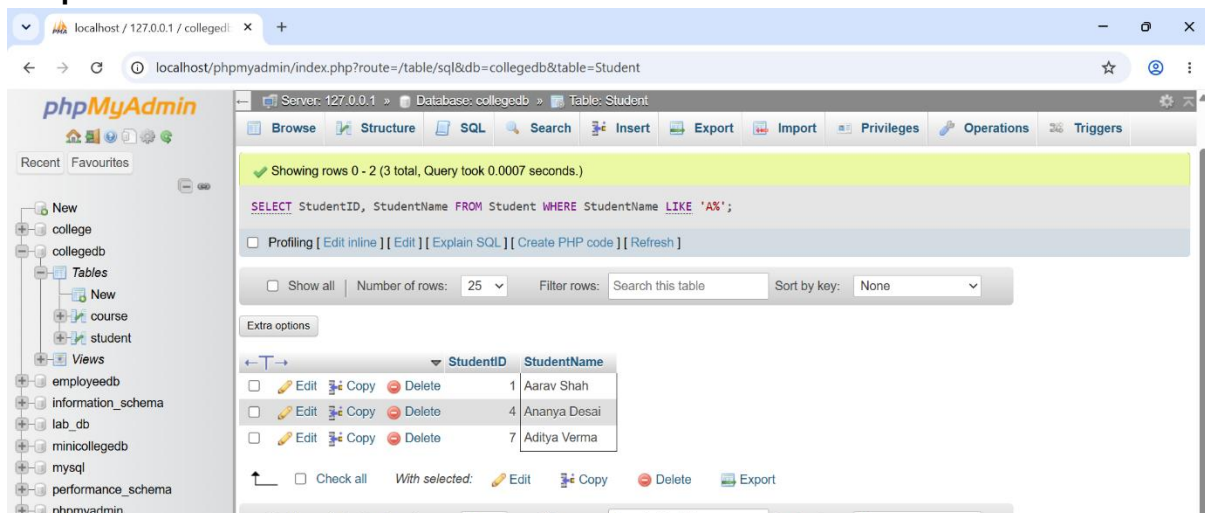
Q5) Write a MySQL query to display students whose names start with the letter A. Use the LIKE operator and display:

Column
StudentID
StudentName

Query-

```
SELECT StudentID, StudentName  
FROM Student  
WHERE StudentName LIKE 'A%'
```

Output-



Showing rows 0 - 2 (3 total, Query took 0.0007 seconds.)

```
SELECT StudentID, StudentName FROM Student WHERE StudentName LIKE 'A%';
```

Number of rows: 25 Filter rows: Search this table Sort by key: None

	StudentID	StudentName
☐	Edit Copy Delete	1 Aarav Shah
☐	Edit Copy Delete	4 Ananya Desai
☐	Edit Copy Delete	7 Aditya Verma

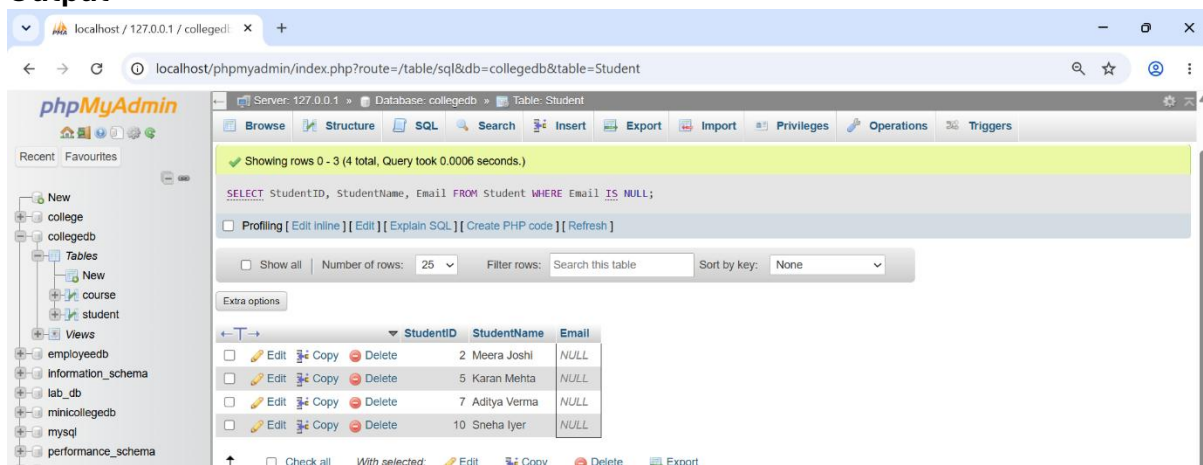
Q6) Write a MySQL query to display students whose email address is missing.
Use the IS NULL operator and display:

Column
StudentID
StudentName
Email

Query-

```
SELECT StudentID, StudentName, Email  
FROM Student  
WHERE Email IS NULL;
```

Output-



The screenshot shows the phpMyAdmin interface. The SQL tab is active, displaying the query: `SELECT StudentID, StudentName, Email FROM Student WHERE Email IS NULL;`. The query executed successfully, showing 4 rows of data. The table structure is as follows:

StudentID	StudentName	Email
2	Meera Joshi	NULL
5	Karan Mehta	NULL
7	Aditya Verma	NULL
10	Sneha Iyer	NULL